

ARYAMAN NAGINENI

Dallas, Texas(Open to Relocate)

📞 312-678-1614 ✉ aryamanrao36@gmail.com 🌐 My GitHub Repository 📁 My Portfolio 🔗 LinkedIn

Professional Summary

Experienced Data Engineer with 5+ years in designing and optimizing scalable data architectures using AWS, Azure, and GCP. Skilled in ETL workflows, real-time data processing, and predictive analytics. Proven track record in enhancing data performance, ensuring compliance, and leading cross-functional teams to drive data initiatives. Strong communicator and committed to driving innovation and operational excellence in data engineering.

Education

University of Illinois at Chicago

Masters of Science in Computer Science

Aug 2022 – May 2024

Chicago, Illinois

Experience

Planstin Administration

Jul 2024 – Present

Senior Data Engineer

Remote

- Developed a high-performance, real-time data streaming platform using Apache Kafka and AWS Kinesis, processing over 2 billion healthcare events monthly, boosting efficiency by 40% and reducing latency by 30%. Modeled a fault-tolerant pipeline with Apache Spark and Amazon S3, featuring cross-region replication for 99.99% uptime. Streamlined Apache Flink for event processing and Elasticsearch for real-time indexing, strengthening analytical capabilities and resilience.
- Revamped data workflows by automating transformations with Apache Airflow, reducing manual effort by 75%. Implemented Terraform to streamline cloud resource provisioning, enhancing the accuracy and timeliness of claims data reporting which Upgraded decision-making and productivity. Delivered Kubernetes for container orchestration to ensure scalable and efficient data job execution across diverse environments.
- Devised sophisticated data models and architectures on AWS and Google Cloud, creating scalable star and snowflake schemas in Snowflake, consolidating ERP, CRM, and transactional data, resulting in a 50% boost in processing efficiency and 35% faster insights, significantly enhancing decision-making. Adopted Apache Hive and Teradata to ensure robust scalability, meeting growing analytics demands.
- Led the creation of a centralized ETL system utilizing AWS Glue, Talend, and Apache NiFi, enhancing data uniformity and accelerating new product launches. Initiated a data quality management system with Deequ and Informatica Data Quality, significantly improving underwriting accuracy and reducing claims discrepancies by over 90%.
- Architected a high-capacity ETL pipeline handling over 5TB of data daily, utilizing PySpark and Databricks for efficient data processing, and Simplified workflow automation and monitoring using Apache Airflow. Refined the infrastructure to support advanced machine learning applications using TensorFlow and Keras for sophisticated fraud detection and customer segmentation.
- Led the migration and software integration of enterprise data systems using AWS (Redshift, S3, Glue, EMR) and Google Cloud (BigQuery, Dataflow). Optimized workflows with Snowflake and Apache Hive, improving query performance by 60% and reducing data costs by 40%. Ensured seamless interoperability between on-premise legacy systems and cloud-native solutions, minimizing downtime during transition.
- Coordinated high-impact projects across IT, analytics, marketing, and finance to enhance data systems integration with business strategies. Led the transition to a modern data platform, boosting operational efficiency by 30% and improving decision-making accuracy by 20%.
- Designed and Distributed a real-time data architecture using Apache Kafka and Apache Flink, processing over 3 billion events daily. Strengthened decision-making speed and accuracy, reducing latency by over 40% and improving customer outcome tracking by 25%, significantly boosting claims delivery efficiency.
- Boosted query optimization in large-scale data marts, improving ad-hoc actuarial and underwriting response times by 70%. Created and Enacted data partitioning and indexing strategies on HDFS and AWS S3, effectively managing high-throughput batch and streaming workloads and maintaining performance during peak loads.
- Engineered advanced analytics solutions using Python, Scala, and Java; Interacted data libraries such as Pandas, PySpark, and NumPy to perform complex data manipulation and analysis tasks. Initiated real-time streaming technologies including Apache Kafka and Spark Streaming in a high-throughput environment, handling over 5TB of data daily.
- Customized visualization solutions using Tableau, Power BI, and Looker to enhance insightful reporting and data-driven decision-making. Customized QlikView dashboards for executive reporting, Improving visibility of key performance indicators across the organization.

University Of Illinois, Healthcare Department

Sep 2022 – June 2024

Data Engineer

Chicago, Illinois

- * Revamped the department's real-time data warehouse, deploying Amazon Redshift, Snowflake, Google BigQuery, and Oracle on AWS and Google Cloud Platform (GCP) infrastructure. Achieved 70% faster query performance and expanded real-time analytics capabilities. Combined Apigee for API management and ElastiCache for Refined caching.
- * Advanced ETL workflow redesign for real-time data processing using Apache Airflow, AWS Glue, Databricks, Dataflow, SSIS, T-SQL, and PL/SQL, reducing processing times by 60%. Leveraged SSIS for orchestrating complex ETL workflows, automating incremental data loads from on-premise SQL Server to cloud data lakes. Improved data integrity with Deequ and advanced operations with Bash automation.
- * Initiated analytical solutions using Tableau and Microsoft Power BI, coupled with predictive analytics models in Python and R, to facilitate data-driven decision-making and strategic insights across various healthcare operations, effectively employing HTML, CSS, and JavaScript to enhance frontend data visualizations. Launched configuration management processes and Employed Apex for aligned programming.
- * Fine-tuned SQL and BigQuery operations, cutting query times by 50% and reducing costs with AWS cost-management. Enriched real-time analytics with predictive analytics and data mining. Contributed DynamoDB for NoSQL data modeling and MFA for security.
- * Instituted stringent data governance protocols to ensure GDPR and HIPAA compliance, Established Apache Atlas for metadata management and AWS Lake Formation to secure data lakes, significantly enhancing data privacy and security measures. Consolidated SSO for user authentication. Strengthened communication skills across multidisciplinary teams, incorporating statistics to better analyze data trends.
- * Orchestrated container orchestration with Kubernetes and infrastructure with Terraform, reducing costs by 40% and enhancing scalability and resilience. Operated Splunk for big data monitoring and analysis. Emphasized DevOps with Kanban to streamline workflow and leveraged Unix for system management. Augmented processing efficiency with Scala and Shell scripting.
- * Expertly configured CI/CD pipelines using Jenkins, CircleCI, and AWS CodePipeline to enhance system reliability and streamline continuous integration processes. Achieved higher deployment frequency and improved stability, significantly boosting operational efficiency.
- * Led cross-functional teams to align real-time data initiatives with business objectives, increasing operational efficiency and revenue by 15%. Accomplished Confluence for documentation and Jira for task management, enhancing project delivery and team collaboration.
- * Facilitated workshops on real-time data engineering and cloud analytics for healthcare professionals, driving innovation and promoting continuous learning. Introduced Infrastructure as a Service (IaaS) solutions to streamline healthcare platforms. Utilized Java and C# to develop advanced healthcare tools and introduced Hadoop, Hive, and HBase to enhance big data processing capabilities.
- * Integrated real-time data pipelines using Apache Beam and Dataflow to enhance healthcare analytics. Calibrated cluster management by leveraging Dataproc for Apache Spark and Hadoop. Deployed scalable microservices with Go, boosting the performance and scalability of real-time healthcare data systems.

- * Developed and implemented scalable data architecture on Microsoft Azure, utilizing Azure Data Factory, Azure Databricks, and Synapse Analytics. Transformed gaming analytics, providing deeper insights into player behavior and game performance. Increased data storage and retrieval with robust data models, leveraging real-time analytics for enhanced player engagement.
- * Enacted the strategic migration of gaming analytics systems to cloud-based platforms, achieving a 40% increase in data processing efficiency. This pivotal move significantly reduced operational costs while enhancing the agility of the data analytics framework to emerging gaming trends.
- * Pioneered and fine-tuned real-time data processing pipelines using Apache Kafka, Spark, and Flink. Managed billions of gaming events per day with minimal latency, dramatically improving real-time analytics of in-game actions and player interactions.
- * Facilitated advanced data analytics and machine learning models in collaboration with the data science team, providing predictive insights. These insights maximized game design decisions, significantly boosting player engagement and retention.
- * Directed the adoption of CI/CD practices for data pipelines using Jenkins and GitLab, markedly improving the deployment frequency of game analytics features. This initiative also ensured higher data quality and reliability, directly contributing to a more robust gaming experience.
- * Championed data governance and compliance initiatives within the gaming sector, ensuring strict adherence to GDPR and CCPA regulations. Engaged data cataloging tools like Apache Atlas and Collibra to maintain high standards of data privacy and security, reinforcing player trust.
- * Played a pivotal role in fostering a data-driven culture within Launchship solutions. This was achieved by advising clients and conducting workshops on best practices in business intelligence and analytics, emphasizing the unique requirements of the gaming industry.
- * Evaluated and Incorporated new data technologies such as Airflow for workflow automation and Docker for containerization, creating a more agile and efficient data engineering environment, essential for rapidly evolving game development cycles.
- * Developed Cassandra and MongoDB for unstructured game data and PostgreSQL for structured data management, supporting multiplayer online games. Expanded scalability and performance using these technologies. Incorporated JSON for data interchange and storage.

Markor Technologies**May 2019 – March 2021**

Data and ETL engineer

Hyderabad, India

- * Spearheaded strategic AWS deployments, enhancing data solutions and processing efficiency with S3, Redshift, EC2, and Glue. Applied IaC, auto-scaling, and load balancing, supporting 25% market reach growth for an e-commerce leader, reducing costs while employing C#.
- * Modernized data pipelines with Apache Airflow, implementing dynamic DAGs and advanced monitoring to streamline workflows and reduce analytics turnaround time by 50%, significantly enhancing data-driven decision-making.
- * Crafted a real-time analytics platform using Amazon Kinesis and Tableau on AWS, boosting customer engagement by 30% and conversion rates. Employed scalable microservices and real-time processing, leveraging machine learning for predictive insights.
- * Aligned AWS IAM and Amazon Macie, establishing robust data security that ensured 100% GDPR and CCPA compliance. Reduced compliance inquiries, enhancing customer trust with automated monitoring and reporting.
- * Spearheaded legacy system migration to AWS, halving processing times and enhancing decision-making speed by 35%. Facilitated AWS Migration Hub and Snowball for efficient transfers, fostering an agile culture through cloud-native adoption. Upgraded data serialization with Parquet for Increased storage efficiency. Managed version control and collaboration using Bitbucket.
- * Applied SOAP web services for seamless communication between heterogeneous systems, enhancing the interoperability and data exchange capabilities of the organization's IT infrastructure.

Projects**SoftCart E-Commerce-IBM Data Platform Development |****October 2024**

- * Architected and executed a hybrid data platform integrating MySQL and MongoDB for operational transactions and catalog data management, enhancing data accessibility and system responsiveness for a leading e-commerce company.
- * Engineered and automated ETL processes using Python and Apache Airflow, which streamlined daily data migrations and supported real-time business intelligence and reporting, reducing data handling time by 30%.
- * Developed a comprehensive data warehousing solution in IBM DB2, implementing advanced SQL analytics features like cubes and rollups that enabled sophisticated reporting capabilities, improving decision-making processes.
- * Led the implementation of real-time analytical dashboards using IBM Cognos Analytics, which provided pivotal sales insights and performance metrics, driving a 20% increase in user engagement through targeted marketing strategies.

Real-time Fraud Detection and Transaction Processing using Advanced Machine Learning Techniques |**January 2023**

- * Orchestrated the deployment of an avant-garde system leveraging Apache Kafka for seamless live data streaming, dramatically amplifying fraud detection efficiency within the financial ecosystem.
- * Innovatively Modeled and Adapted tailor-made ML algorithms, achieving an extraordinary 95% success rate in pinpointing fraudulent transactions, substantially mitigating false positives.
- * Masterful navigated the management of over 100 million transactions daily via Apache Spark, securing ultra-low latency in fraud detection while upholding stellar system performance.
- * Employed a comprehensive array of datasets to empirically demonstrate a machine learning model's capacity to diminish fraud by 60%, thereby potentially safeguarding assets in excess of \$500 million, exemplifying the project's scalability.

Scalable Data Pipeline Construction and Optimization for Real-time Analytics |**November 2022**

- * Unified EHRs, wearables, and social determinants using Apache NiFi and Talend, incorporating real-time analytics and blockchain for secure, interoperable data sharing, enhancing data integrity and responsiveness.
- * Performed HL7 FHIR protocols with smart contracts to automate and ensure consistent data exchanges, streamlining healthcare operations and enhancing ecosystem efficiency.
- * Engineered predictive models with TensorFlow, augmented by federated learning for decentralized, privacy-preserving analytics, personalizing healthcare management while safeguarding patient data.
- * Sketched a user-friendly interface with React.js and D3.js, integrating advanced data visualization like interactive 3D models for accessible, engaging health data insights and Streamlined patient outcomes.

Technical Skills**Languages:** Python, Java, C, c++, HTML/CSS, JavaScript, SQL, R, Scala, Bash**Developer Tools:** Apache Kafka, Apache Spark, Apache Flink, Apache Airflow, PySpark, Jenkins, CircleCI, AWS CodePipeline, Terraform, Tableau, Power BI, Looker, QlikView, Git, GitLab, Splunk, Visual Studio Code, Eclipse, Android Studio, Databricks, Snowflake, IBM Cognos Analytics, MySQL, MongoDB**Technologies/Frameworks:** AWS (Redshift, S3, Glue, EMR, IAM, Lambda, Kinesis), Google Cloud Platform (BigQuery, Dataflow, Dataproc, GCP infrastructure), Microsoft Azure (Azure Data Lake, Synapse Analytics), Kubernetes, Docker, Apache Hive, HDFS, Hadoop, TensorFlow, Keras, Informatica Data Quality, Deequ, Oracle Database, DynamoDB, Redis/ElastiCache, PostgreSQL, Cassandra, MongoDB, HL7 FHIR Protocols, React.js, D3.js, JSON, SOAP Web Services, REST APIs**Publications** :link: <http://doi.org/10.1729/Journal.28537>